How Predictable Is Brazilian Aviation?

Machine Learning for Airport Traffic, Seat Occupancy, and Route Demand

Author: Caio Bandeira · **Project:** ANAC Flight Intelligence v2 · **Evidence:** 2023 validation, out-of-sample

> **Scope of every number in this report.** All results are **validation results on 2023**, data the models did not see during training or early stopping. They are **not** results of the final test (2024–2025), which has **not been run** (Section 6.4), and they are **not** live forecasts: no model serves predictions in production. Every figure and table is regenerated by `research/ml-showcase/reproduce.py` and `analysis.py` from the saved artifacts.

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1. Executive Summary

Three forecasting problems were built on 26 years of ANAC flight-leg records, each compared against simple baselines on 2023 data that played no part in fitting:

Model	What it forecasts	Best simple baseline	LightGBM	Verdict
A	Daily takeoffs + landings at the 50 busiest airports, 1 and 3 months ahead	Recent same weekday: WAPE **10.30%**	WAPE **10.10%**	**No conclusive gain.** Difference −0.21 pts, 95% interval −0.84 to +0.35. The baseline is the method of record.
B	Paid load factor per route and carrier, 1, 3 and 6 months ahead	Mean of last 12 known months: **6.54 pts**	**6.06 pts** seat-weighted MAE	**Beats the baseline**, −0.48 pts (−0.94 to −0.04). Conclusive overall and at 6 months; not on its own at 1 or 3 months.
D	Monthly paid passengers and seats per route, 1, 3 and 6 months ahead	Same month last year: WAPE **25.31%** (pax), **25.73%** (seats)	WAPE **22.24%** (pax), **22.40%** (seats)	**Beats the baseline** overall, −3.07 and −3.33 pts. **But** it under-forecasts by about **12%** and is **no better than the baseline at 6 months**.

What this means.

- Airport traffic is highly predictable from its own recent rhythm (about 10% WAPE), and a gradient-boosted model adds nothing measurable on top of a simple weekday rule.
- Seat occupancy is predictable to about 6 points, and the model's gain is real but small.
- Route demand is where machine learning adds the most at short horizons: at 1 month ahead its WAPE is 18.7% against 24.1% for the best baseline.
- Route demand is also where the model fails most clearly. It inherits the level of the year before and missed 2023's 15% market growth, so it should not be used without a level correction.

What was not done, on purpose.

- No hyperparameter search.
- No test-set evaluation.
- No SHAP: feature attributions are LightGBM gain importances, labelled as such.
- No user-input "prediction playground" (Section 14.3 explains why).

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2. Research Questions

1. **RQ1 — Predictability.** How accurately can airport movements, seat occupancy and route demand be forecast 1 to 6 months ahead from public ANAC data alone?
2. **RQ2 — Value of ML.** Does a gradient-boosted model beat simple, transparent baselines by more than chance?
3. **RQ3 — Where it fails.** Which airports, routes, carriers, volumes, horizons and months concentrate the errors, and is the error biased?
4. **RQ4 — Use.** Given the measured error, what can these forecasts responsibly support?

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3. Dataset & Data Quality

Source. ANAC's public statistical air transport microdata, one row per flight leg, from 2000-01 to 2025-12. That is a ~20 GB CSV, ingested by an audited, resumable pipeline into the `flights` Parquet dataset. The raw file is never modified.

Quality controls that shape the results:

Issue	Treatment
Four months have no data: 2015-08, 2016-02, 2018-07, 2020-08	Treated as **unknown, never zero.** Lags falling on them are null, and means divide by known months only.
Passenger scope	Flights that took off with seats, excluding positioning, zero-distance and same-airport legs.
Load factor above 1.5	Treated as a recording error for B and D (12,710 legs across all years). Values from 1 to 1.5 are kept.
Per-leg counting	A passenger on a two-leg flight counts twice. Totals run about 1–2% above ANAC's published figures.
No fares	The data has no ticket prices, so no price-related target exists.

Model datasets (target months; the test period is not written to disk):

Module	Unit	Train	Purged	Validation	Inputs
A	Airport × day, top 50 airports	356,050 rows (2013-01 → 2022-10)	9,150	36,500	21 (10 calendar, 6 history, airport, timing)
B	Route × carrier × month	488,226 (2013-01 → 2022-10)	15,591	47,197, of which 42,369 flew and were scored	42 (29 history, 6 calendar, origin, destination, carrier, distance, …)
D	Route × month	318,149 (2013-01 → 2022-10)	11,052	33,762	38 (26 history, 6 calendar, origin, destination, distance, …)

In B, about 90% of eligible units flew in the target month. Its errors are **conditional on the unit flying**: the model does not predict whether a route is cancelled.

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4. Exploratory Data Analysis

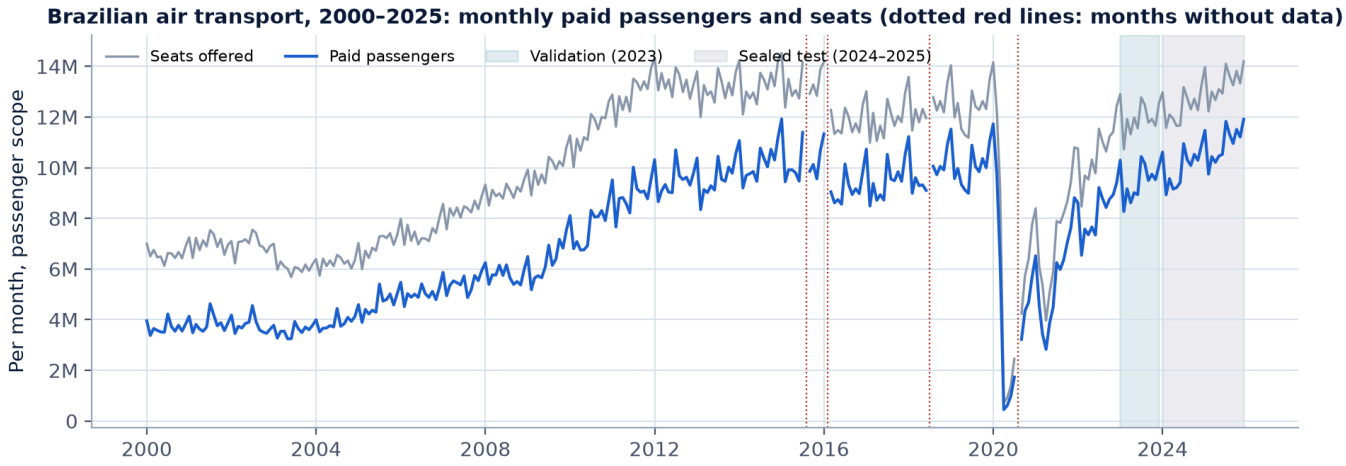


Figure 1. Paid passengers and seats per month, 2000–2025. Gaps are the four absent months. The market totals for 2024–2025 come from the analytical layer that the dashboard already shows; ***no model saw or was evaluated on them.***

- **Long-run growth** to a 2019 peak of 120.7 M paid passengers, a collapse in 2020, and a recovery through 2021 (67.8 M), 2022 (98.6 M) and 2023 (113.7 M).
- **Validation sits in an unusual year.** 2023 grew **15.3% over 2022**. Any method anchored on last year's level starts behind: this is the single most important fact for reading Model D's results.
- **Strong annual seasonality:** January and July peaks (holidays), February and the mid-year lows.

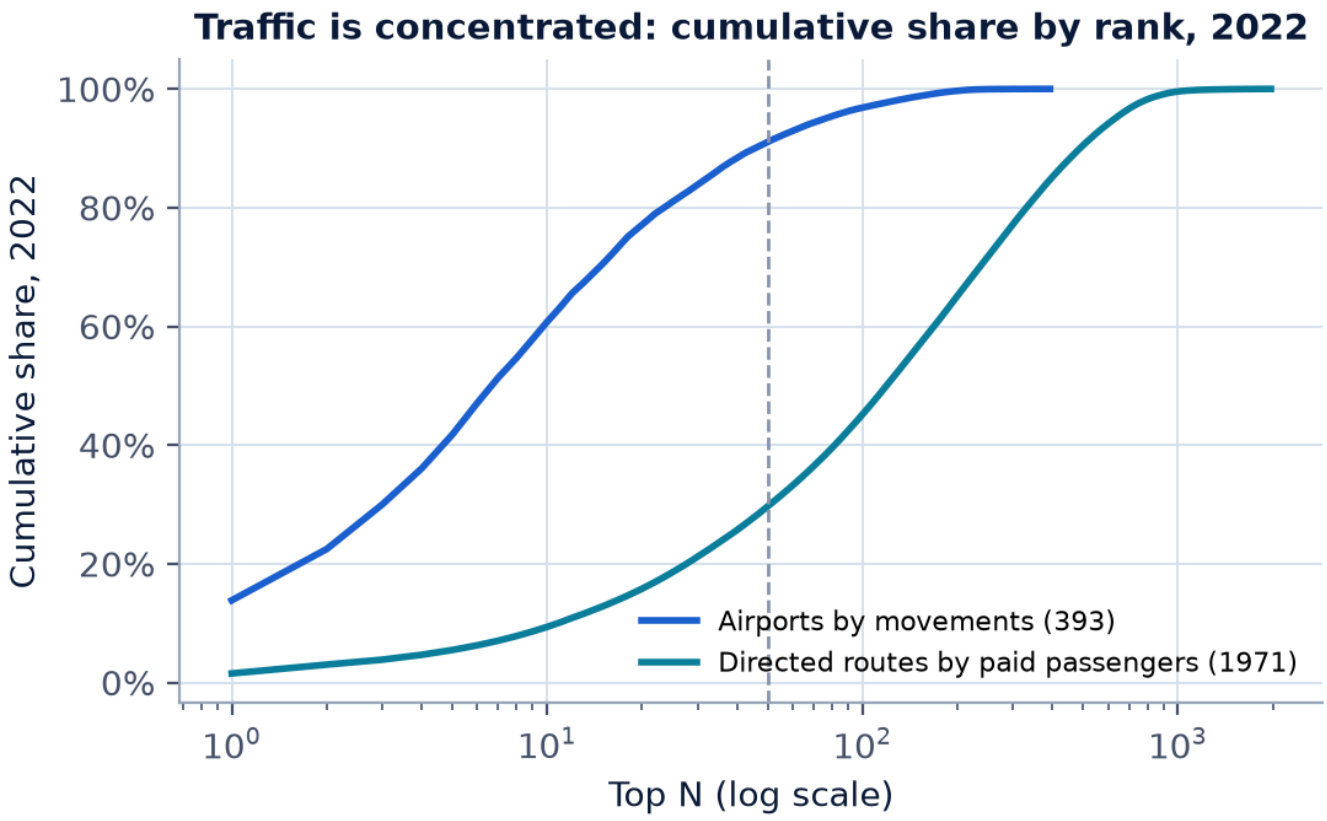


Figure 2. Cumulative share by rank, 2022.

- **Airports:** the top 50 of 393 handled **91.1%** of all aircraft movements, which is why Model A covers them.

- **Routes:** traffic is spread more widely. **331** of 1,971 directed routes carry 80% of paid passengers.

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5. Feature Engineering

Every feature is **point-in-time**:

1. At the start of origin month O, the last month known is $C = O - 2$, because publication lags by about two months.
2. The target is $T = O + h$.
3. Feature functions receive history already cut at C and raise `LeakageError` if a later row reaches them.
4. Targets are built in separate functions and stored in separate files.
5. Tests scramble or delete everything after the cutoff and require identical features (`tests/test_features_leakage.py`, `tests/test_real_features.py`).

Group	Examples	Used by
Calendar	day of week, week of year, month, school vacation, holidays, Carnival, days in month, weekend days	A, B, D
History (level)	recent 28/56-day level, same weekday mean over 8 weeks, means over 3/6/12 months	A, B, D
History (seasonal)	lag 364/728 days; lag 12/24 months; year-over-year change	A, B, D
Load factor history	route, carrier, origin and destination load factor; volatility	B, D
Structure	seats per flight, carriers on route, months observed, unknown months in window	B, D
Identity	airport; origin, destination, carrier; distance; international flag	A, B, D

Absolute time (the year) is excluded on purpose, so trees cannot memorise a calendar period.

Scale-free targets. Volumes (A, D) are modelled as a ratio to the unit's recent level, because trees cannot extrapolate. With L1 loss weighted by that level, $|y - p \cdot \text{level}| = \text{level} \cdot |y/\text{level} - p|$. The model therefore minimises WAPE directly.

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6. Experimental Design

6.1 Splits

Split	Target months	Role
Train	2013-01 → 2022-10 (h = 1); earlier for longer horizons	Fitting
Early stopping	Targets from 2022-01 within training	Choosing the number of trees
Purge	Targets not yet known at the first validation origin	Discarded, to prevent look-ahead
Validation	**2023-01 → 2023-12**	**All results in this report**
Test (sealed)	2024-01 → 2025-12	Not built, not read, not run

6.2 Protocol fixed in advance

- **One configuration.** A single conservative LightGBM configuration (`configs/models.toml`), fixed before any validation result was seen. No hyperparameter search.
  - Settings: `objective = l1`, `learning_rate = 0.03`, `num_leaves = 31`, `min_data_in_leaf = 200`, `feature_fraction` and `bagging_fraction` 0.8, `lambda_l2 = 5`, `seed = 42`, `deterministic = true`.
- **Selection rule, fixed in advance.** A model replaces its best baseline only if the 95% interval of the paired error difference excludes zero.
- **Uncertainty.** Intervals come from a **block bootstrap over the 12 validation months**, so they respect within-month correlation.
- **Freezing.** After validation, each model was refit on everything known at the first test origin, with the same number of rounds, and frozen in `models/v1` with SHA-256 hashes. `test_period_used: false` is recorded in each manifest.

6.3 Reproduction (this study)

Check	Result
Metrics recomputed from the saved validation predictions with the project's own `compare()`	Identical to `reports/models/*.json` (difference < 1e-12) for A, B, D-pax and D-seats
Selection models **refit from scratch in memory** with the frozen configuration	**Bit-identical predictions** (maximum absolute difference 0.0) and the same best iteration (98, 46, 395, 243)
SHA-256 of `models/v1/*/*.txt` against the manifests	All four match
Environment	Python 3.12.10, LightGBM 4.7.0, Polars 2.1.0, NumPy 2.5.4

Full record: `research/ml-showcase/results/reproduction.json`.

6.4 Why the final test was not run

The brief allowed one final evaluation on 2024–2025 only with evidence that the period stayed untouched. That evidence is incomplete:

1. **Decision 3** in `docs/decisions.md` records that annual totals for 2024–2025 were seen during the data audit. The test period is therefore **not fully blind**.
2. The project's own plan (`docs/models.md`, known issue 6) is to retrain on a dataset with the four missing months recovered, freeze as `models/v2`, and then test. Testing `v1` now would spend the test period on models already scheduled for replacement.
3. The 2024–2025 market totals are visible in the dashboard and in Figure 1. Any further look at that period, including this study, makes it less blind.

The test was therefore **not executed**. The frozen models and their hashes are ready for the test once the owner unseals it under the documented procedure.

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7. Models & Baselines

Module	Baselines (all computed from data known at the cutoff)	Model
A	Same weekday a year ago; same, with trend; **recent same weekday (mean of the last 8 same weekdays)**	LightGBM on the ratio to the 56-day level
B	**Mean of the last 12 known months**; same month last year; mean of the last 3 months	LightGBM on load factor, L1 weighted by seats
D	**Same month last year**; same, with trend	LightGBM on the ratio to the 12-month mean, one model for passengers and one for seats

8. Evaluation Metrics

Metric	Definition	Why
WAPE	$\Sigma\lvert\hat{y} - y\rvert / \Sigma\lvert y\rvert$	Scale-free for volumes, robust to small units, interpretable as "share of traffic misplaced"
Seat-weighted MAE	Σ seats$\cdot\lvert\hat{y} - y\rvert / \Sigma$ seats, in percentage points	Load factor is already a ratio; weighting by seats reflects capacity
Bias	$\Sigma(\hat{y} - y) / \Sigma\lvert y\rvert$	Systematic over (+) or under (−) forecasting
RMSE	$\sqrt{\text{mean}((\hat{y} - y)^2)}$	Penalises large misses (A, D only)
95% interval	Block bootstrap over 12 months	Uncertainty of each metric and of paired differences

9. Results — Model A: Airport Traffic

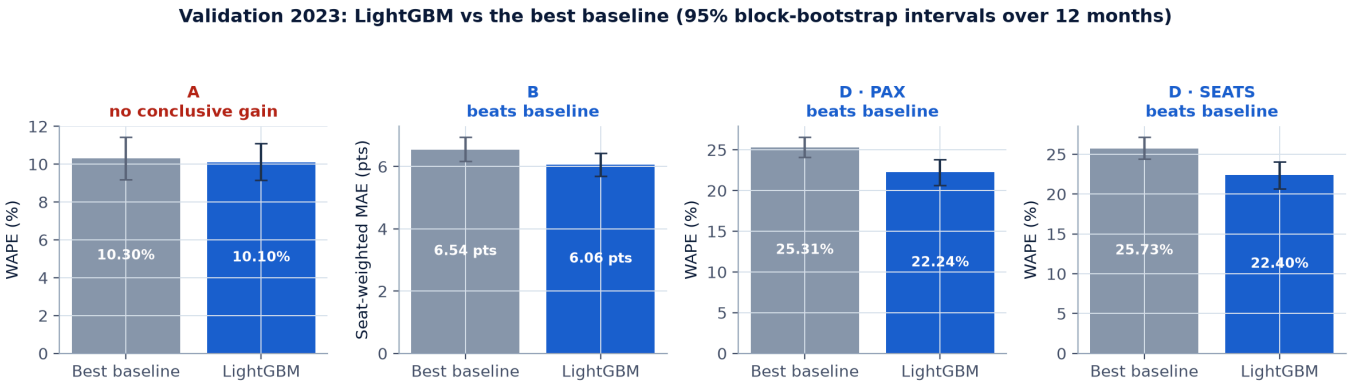


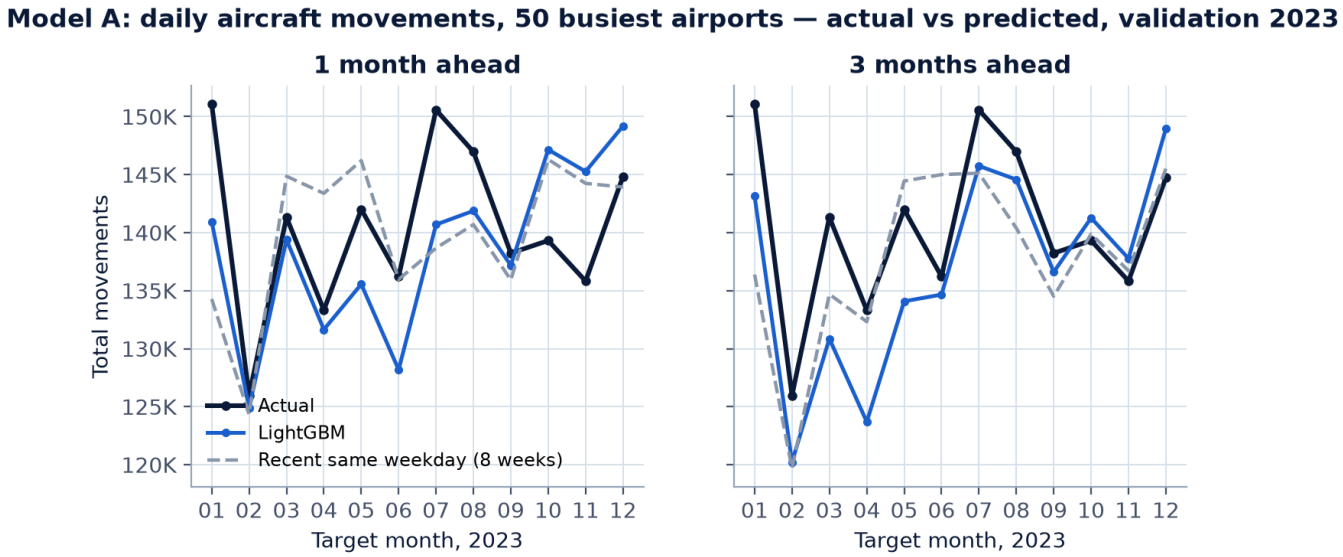
Figure 3. All modules: LightGBM against the best baseline, with 95% intervals.

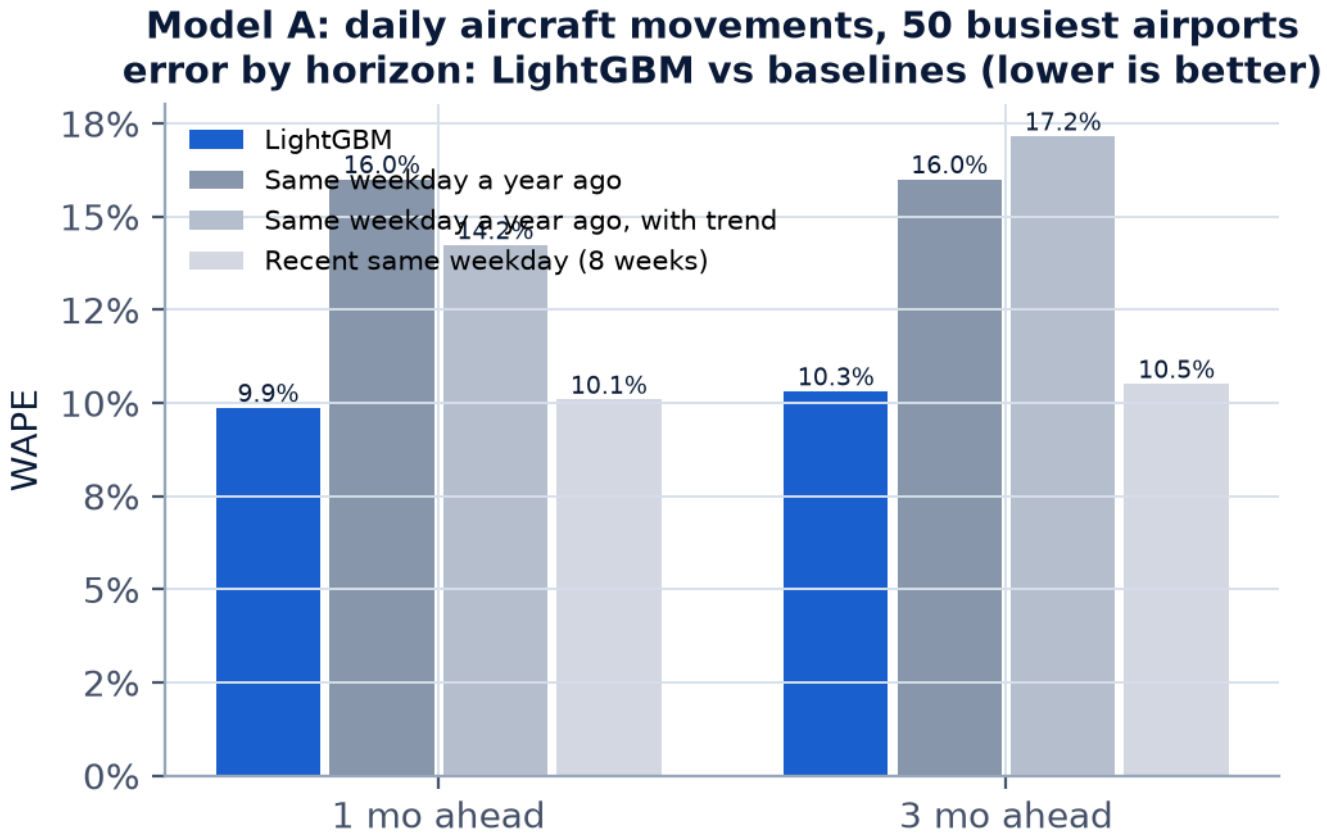
Method	WAPE	95% interval
LightGBM	**10.10%**	9.15–11.10%
Recent same weekday	10.30%	9.19–11.43%
Same weekday a year ago, with trend	15.70%	14.63–16.79%
Same weekday a year ago	16.00%	14.43–17.74%

Paired difference: −0.21 pts (−0.84 to +0.35), which is **not conclusive**. **Bias:** −2.0% (−3.8% to +0.1%).

Horizon	LightGBM WAPE	Best baseline	Difference (95% interval)	RMSE, LightGBM / baseline
1 month	9.87%	10.09%	−0.22 (−0.96 to +0.46)	18.45 / 18.22 movements
3 months	10.32%	10.51%	−0.20 (−0.87 to +0.47)	19.19 / 18.87 movements

The model's WAPE is slightly lower, but its **RMSE is slightly higher**. It trades a little average accuracy for somewhat larger misses.





Interpretation. Daily airport traffic follows a stable weekly schedule. A rule that copies the recent same weekday captures almost all of it, and 98 trees on the same information add nothing measurable. **The baseline is the method of record for Module A.**

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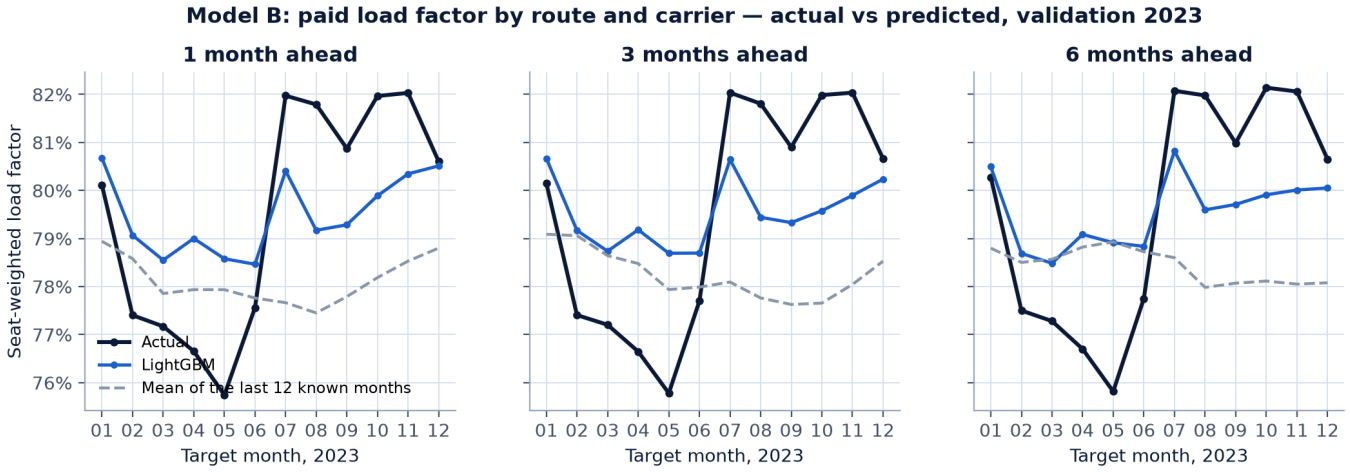
10. Results — Model B: Seat Occupancy

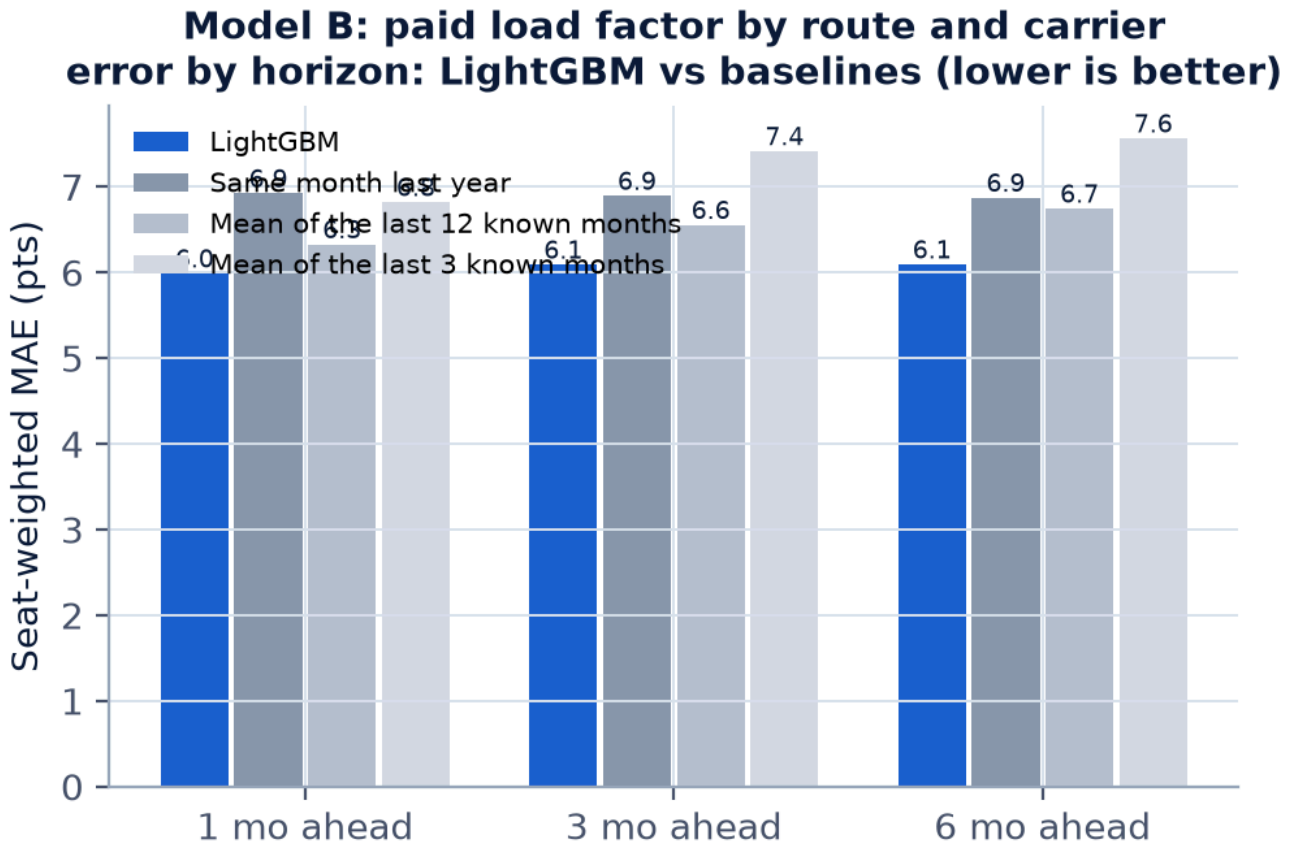
Method	Seat-weighted MAE	95% interval
LightGBM	**6.06 pts**	5.70–6.43
Mean of last 12 known months	6.54 pts	6.17–6.95
Same month last year	6.90 pts	6.51–7.30
Mean of last 3 months	7.27 pts	6.93–7.57

Paired difference: −0.48 pts (−0.94 to −0.04), which is **conclusive**. **Bias:** +0.04% (−0.91% to +1.02%), essentially unbiased.

Horizon	LightGBM	Best baseline	Difference (95% interval)
1 month	5.99	6.33	−0.34 (−0.83 to +0.15)
3 months	6.09	6.56	−0.46 (−0.95 to +0.02)
6 months	6.09	6.75	**−0.65 (−1.08 to −0.28)**

The gain grows with the horizon. At 1 and 3 months the interval still includes zero, so only the pooled result and the 6-month result are conclusive.





11. Results — Model D: Route Demand & Capacity

Target	LightGBM WAPE	Same month last year	With trend	Difference vs best (95% interval)	LightGBM bias
Paid passengers	**22.24%** (20.65–23.84)	25.31%	30.52%	−3.07 (−4.97 to −1.09)	**−12.2%** (−15.5 to −8.9)
Seats	**22.40%** (20.69–24.04)	25.73%	32.57%	−3.33 (−5.32 to −1.19)	**−11.8%** (−15.3 to −8.3)

Horizon	Pax: LightGBM vs best baseline	Seats: LightGBM vs best baseline
1 month	**18.74% vs 24.07%** (−5.33; −7.06 to −3.72)	**18.71% vs 25.72%** (−7.01; −7.99 to −5.99)
3 months	**21.31% vs 25.35%** (−4.04; −5.90 to −2.09)	**21.49% vs 25.76%** (−4.28; −6.11 to −2.40)
6 months	26.71% vs 25.09% (**+1.63**; −0.61 to +4.25)	27.04% vs 25.53% (**+1.51**; −0.94 to +4.61)

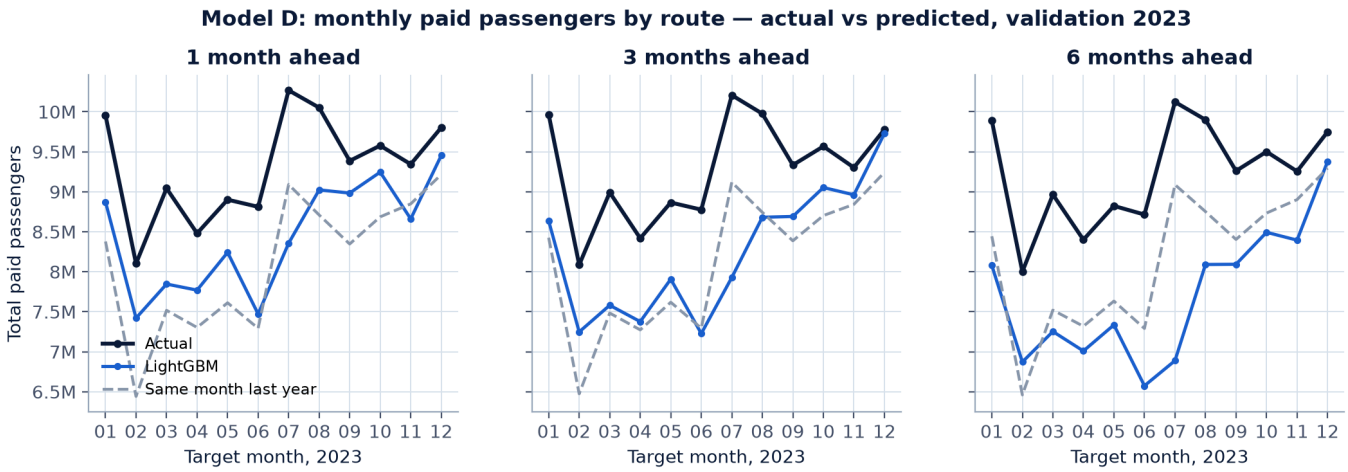
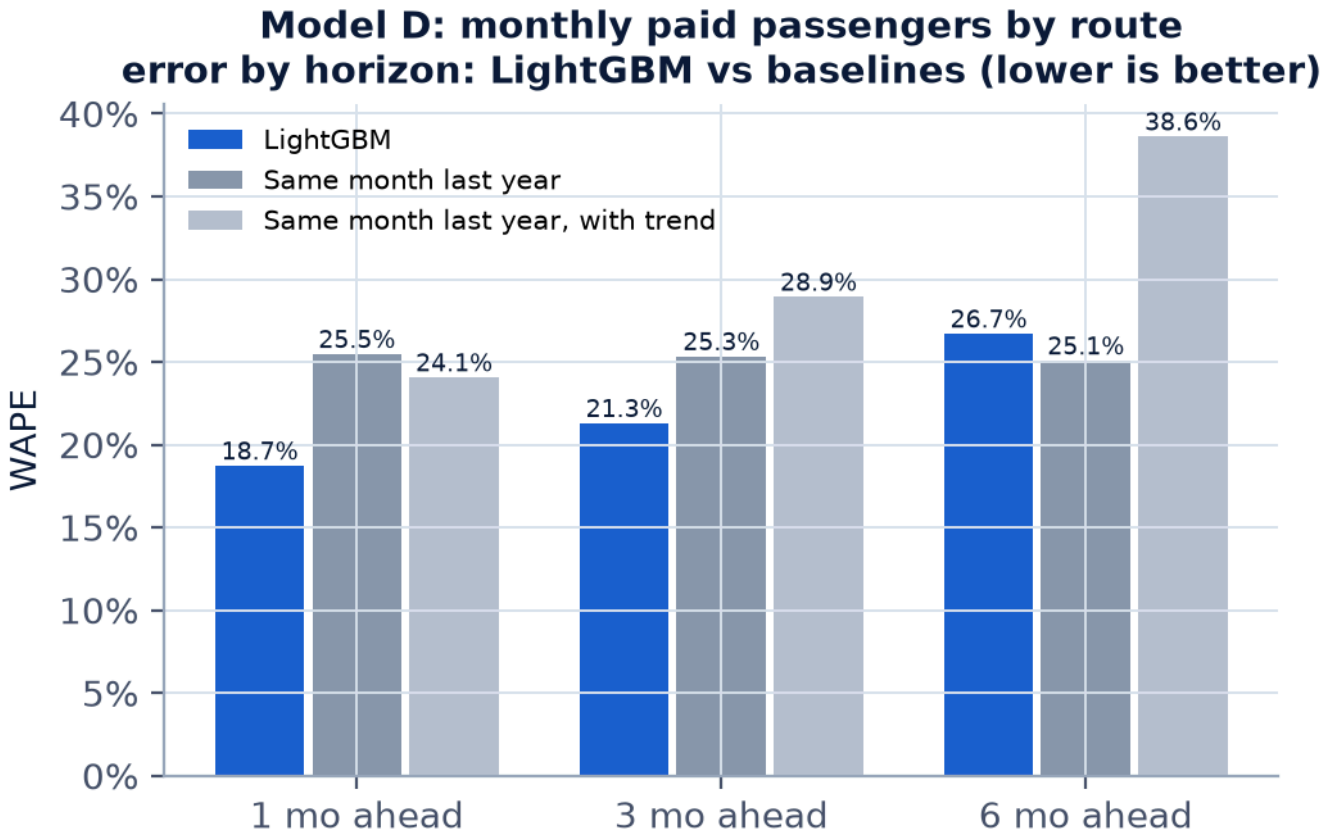
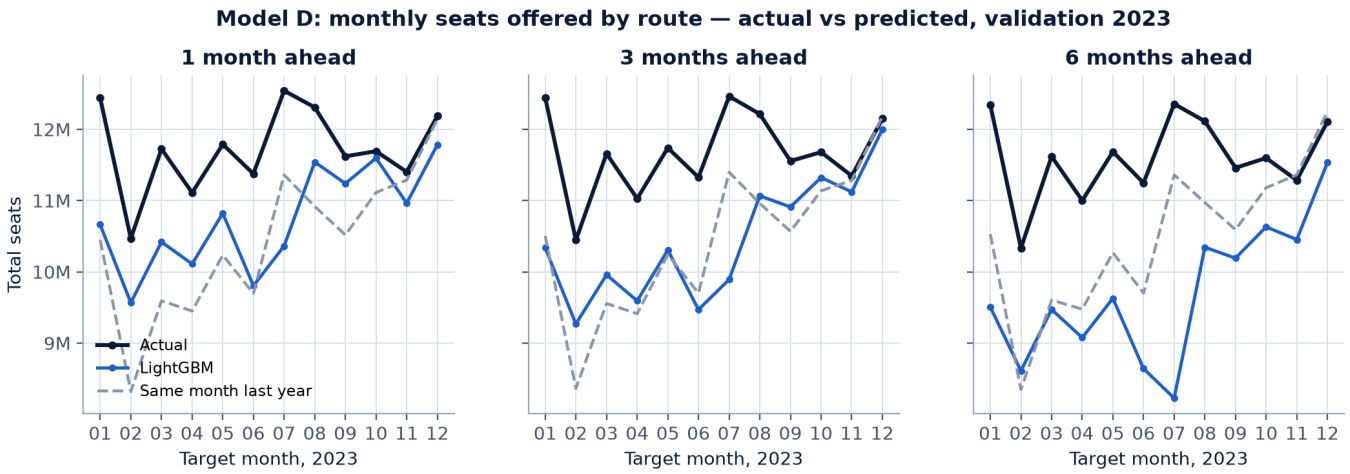


Figure 4. Total paid passengers on modelled routes. LightGBM (blue) tracks the shape of the year better than last year's values (dashed) but stays below the actual level (black) in every month and at every horizon.

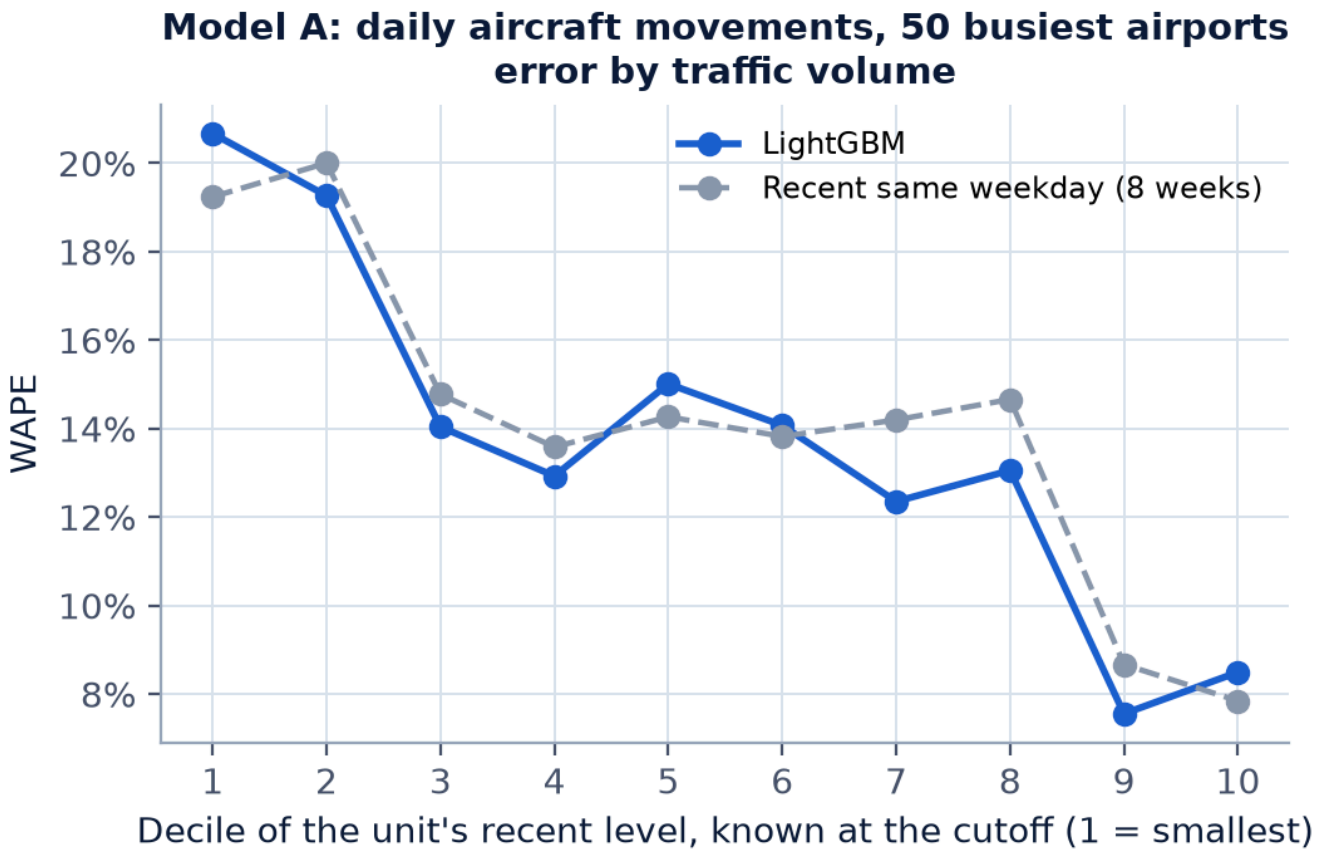
- **Bias by horizon:** about −9% at 1 month, −11% at 3 months and −16% at 6 months.
- **The 6-month model is no better than copying last year.** Only the 1- and 3-month models add value.

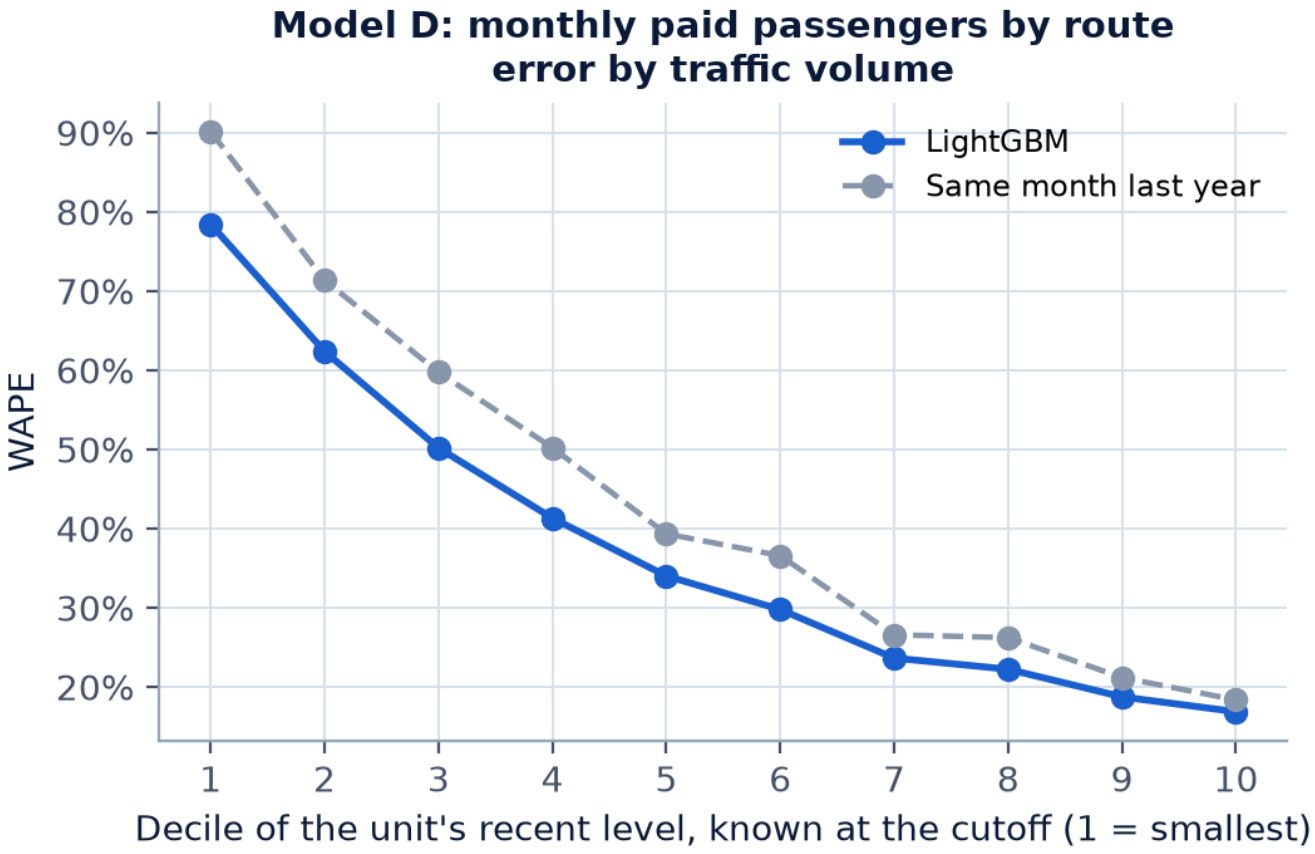




12. Error Analysis

12.1 Error by traffic volume

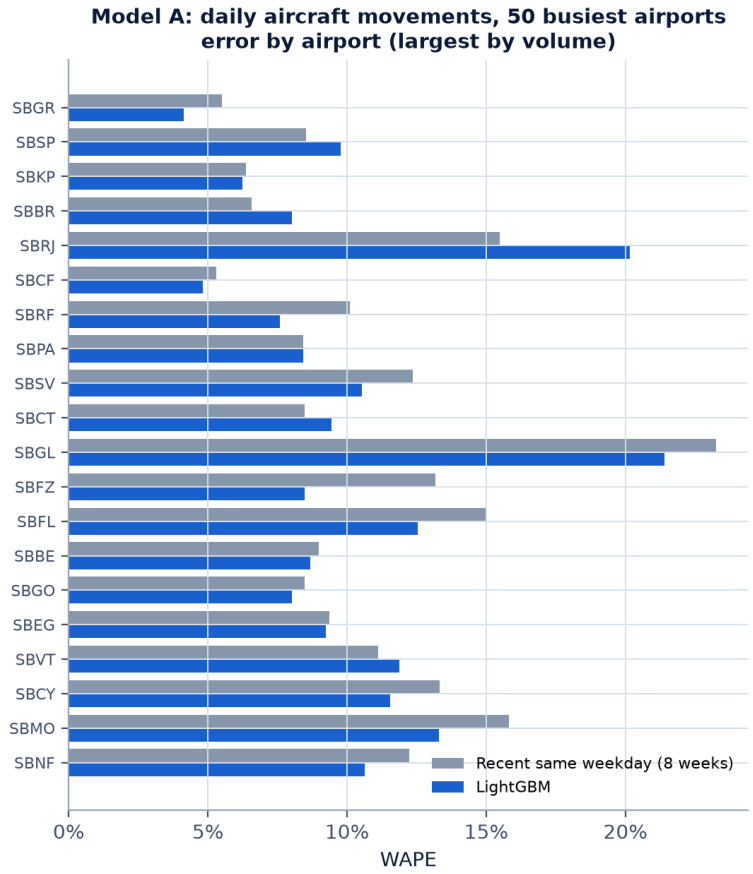




Deciles use the unit's level **known at the cutoff**, never the target.

- **A:** WAPE falls from about 20% for the smallest of the top 50 airports to 8–9% for the largest. The model and the baseline swap places from decile to decile, which is consistent with no real difference between them.
- **D:** error falls steeply with route size, from 78% (LightGBM) and 90% (baseline) in the smallest decile to 17% and 18% in the largest. **LightGBM is better in all ten deciles**, and most of its gain comes from small and medium routes.
- **B:** error falls from 9.3 to 5.4 pts with capacity, and the model leads in 8 of 10 deciles.

12.2 Error by entity



- **A, best:** Guarulhos (SBGR) has a WAPE of 4.1% against 5.5% for the baseline.
- **A, worst:** Santos Dumont (SBRJ) has 20.2% against 15.5%.

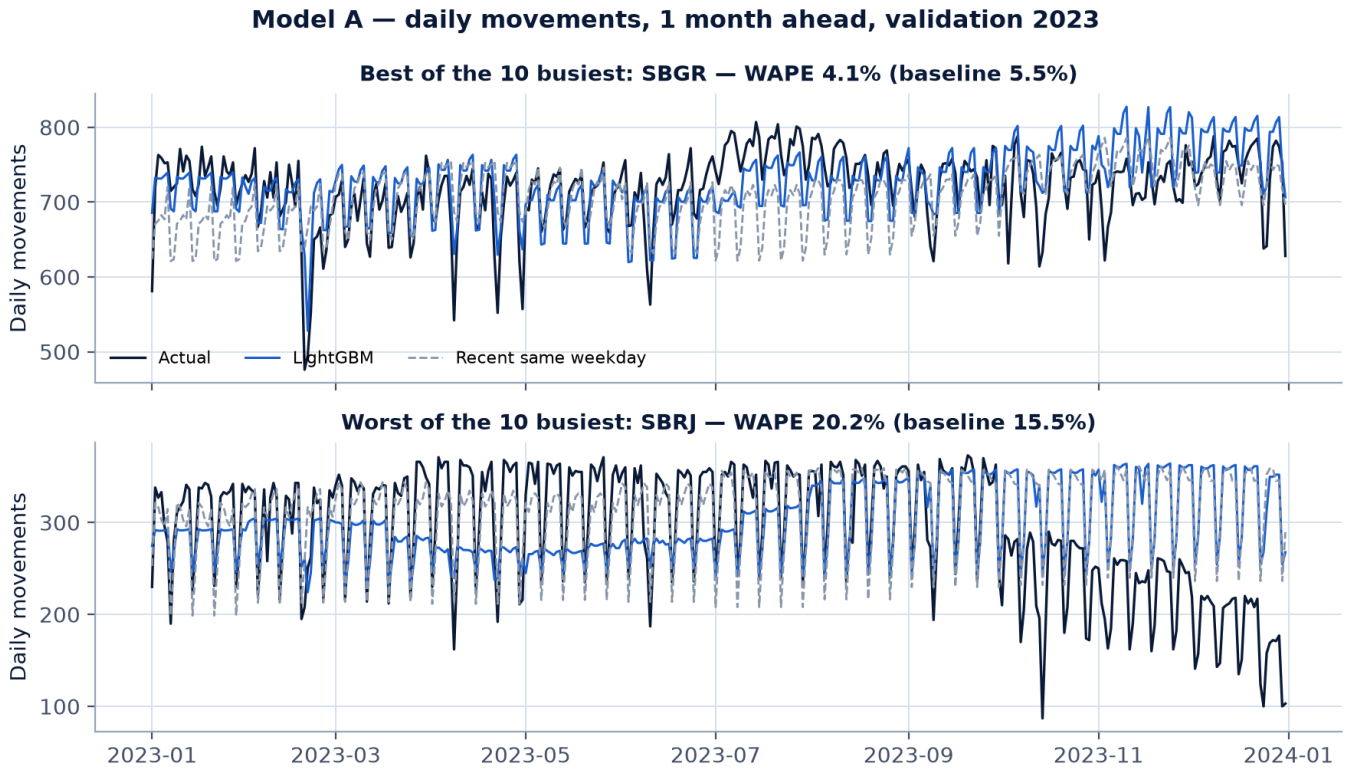
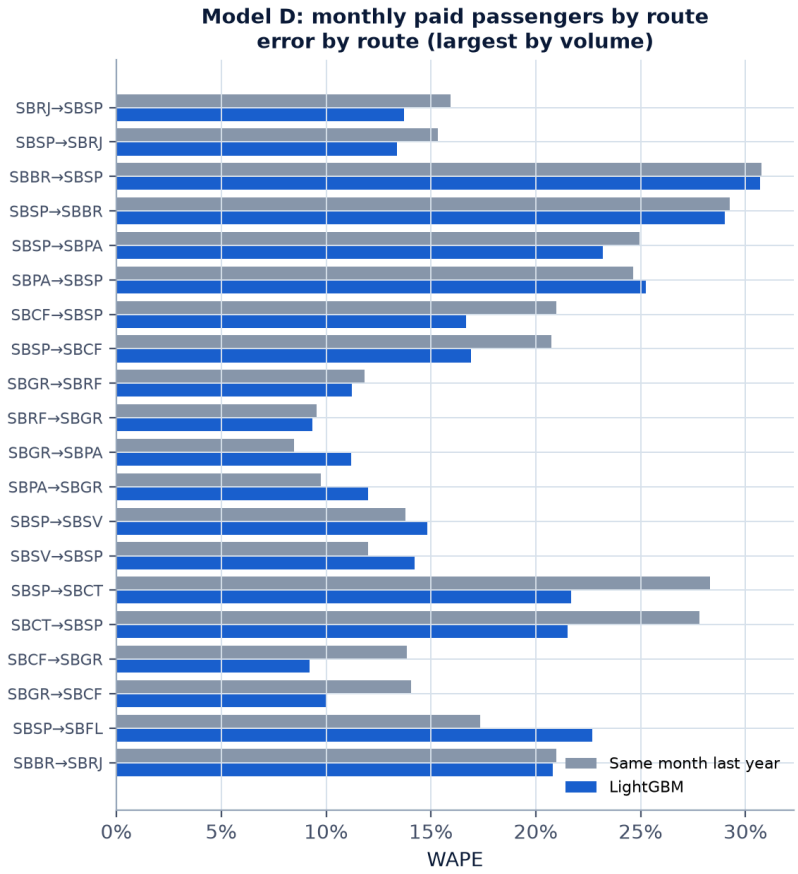


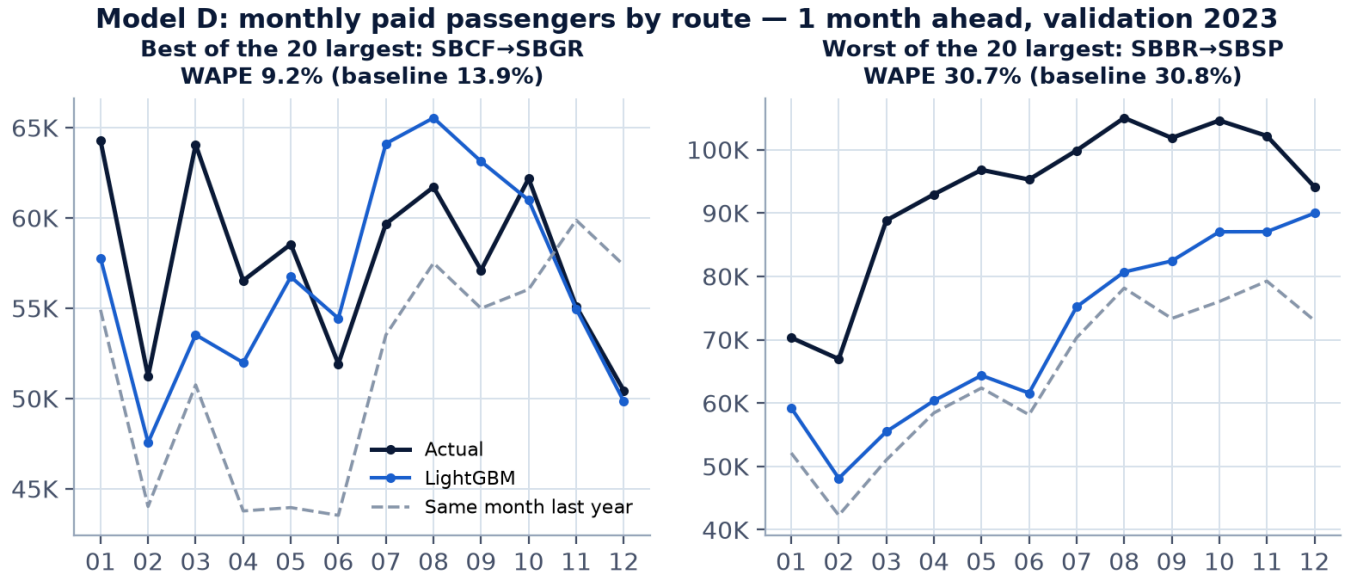
Figure 5. Daily movements, 1 month ahead. At SBRJ, traffic drops sharply from October 2023: monthly movements go from 9,678 in September to 7,514 in October and 5,584 in December. Neither method anticipated this structural break, which history alone cannot reveal. The model also under-predicted SBRJ in the first half of the year.



D, trunk routes. On several of the largest routes, LightGBM and the baseline under-forecast by almost the same amount (around −30% on Brasília–Congonhas). On these routes both methods fail together.

Route	Growth 2022 → 2023 (paid passengers, both directions measured separately)
Brasília → Congonhas	+44%
Congonhas → Porto Alegre	+33%
Santos Dumont → Congonhas	+19%

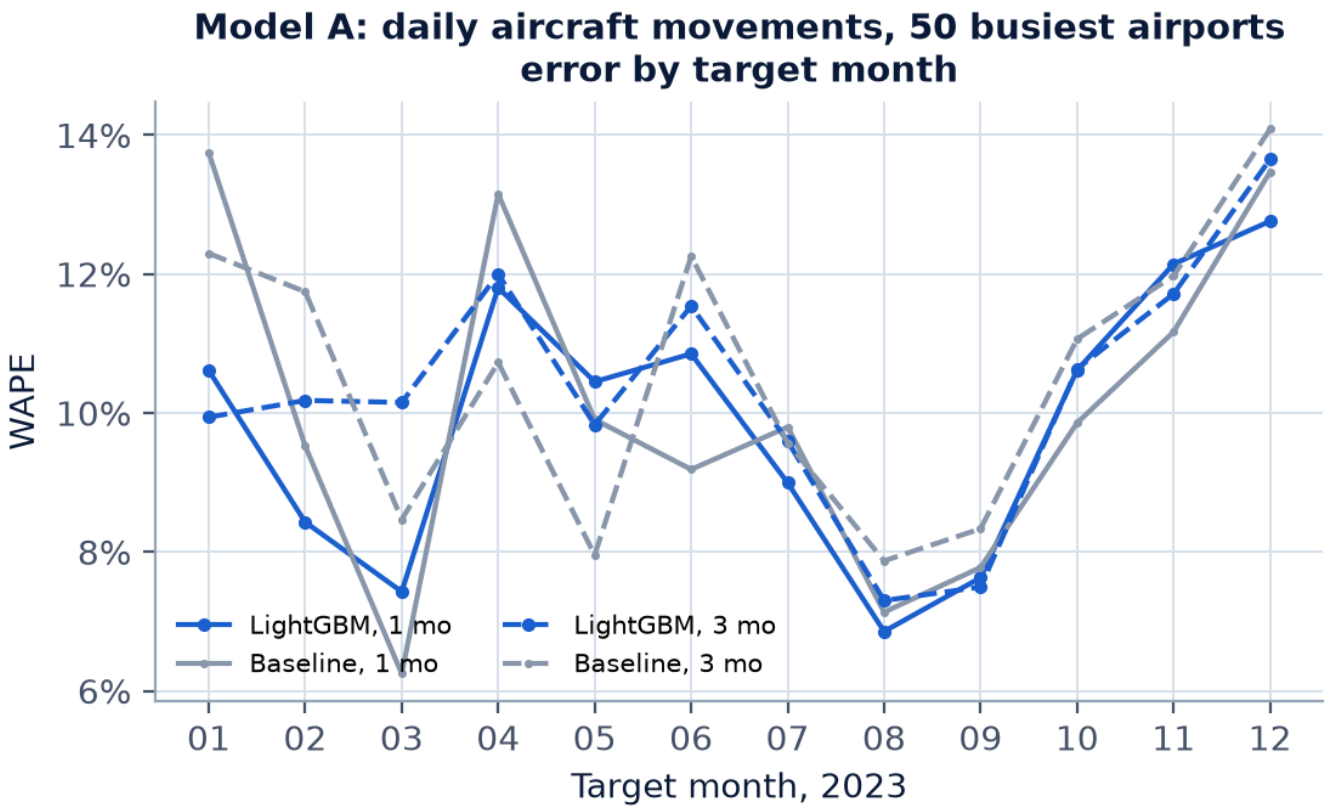
Reading: the 2022 history did not contain this growth, so no method anchored on it could predict it. This is a measured coincidence of growth and bias, not a proven cause.

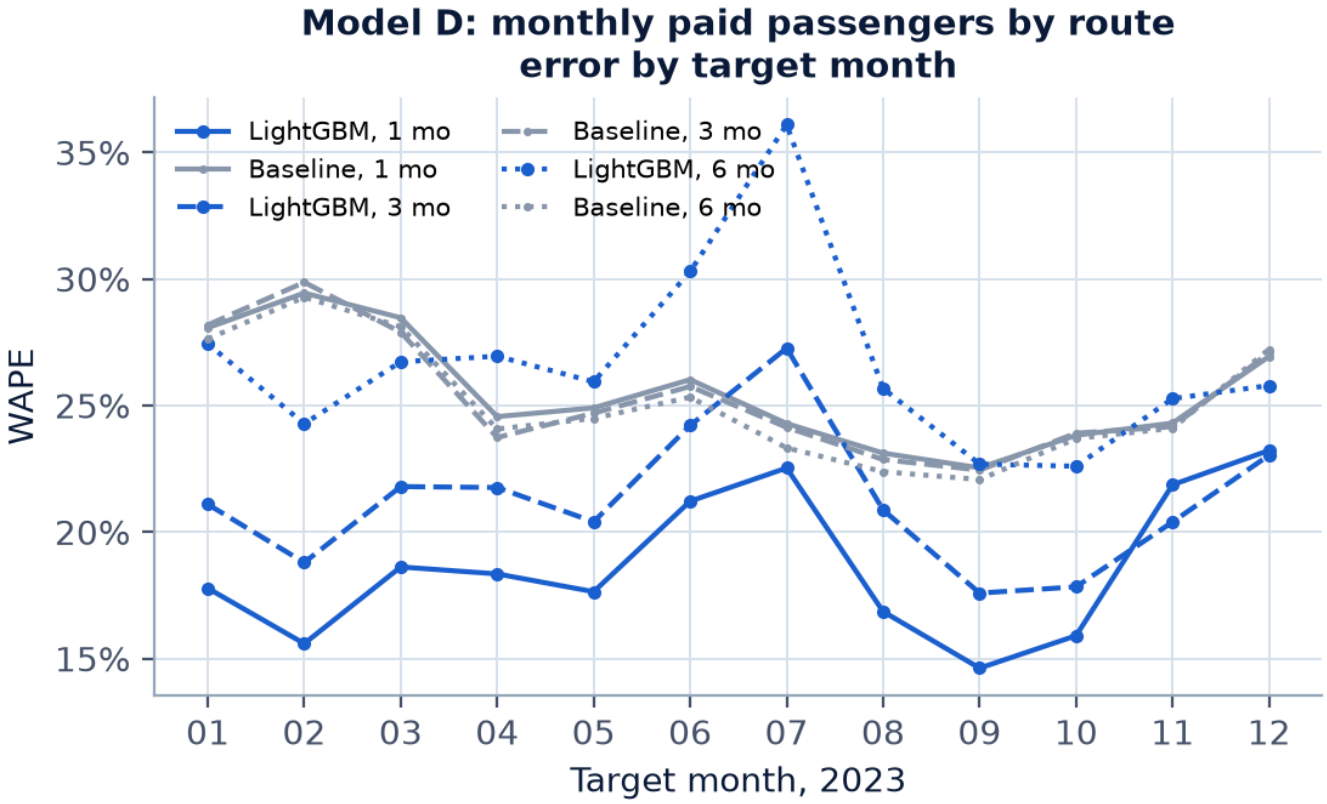


B by carrier and market.

- **Carriers:** the three largest domestic carriers (TAM, GLO, AZU) improve by 0.4–1.0 pts over the baseline. Some foreign carriers, such as LAN and CMP, are better served by the 12-month mean.
- **Domestic** routes improve (5.83 vs 6.55 pts). **International** routes do not (7.12 vs 6.50 pts).

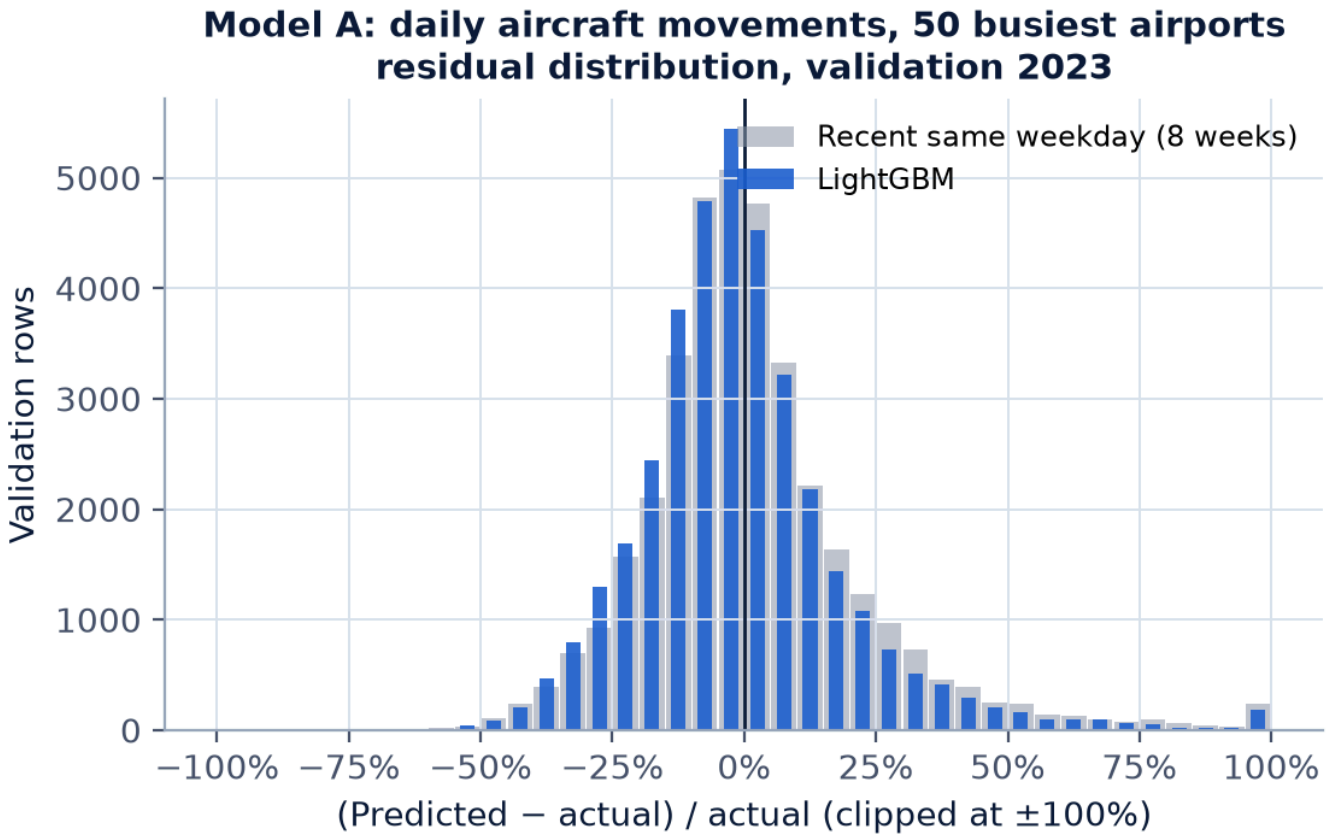
12.3 Error by month and weekday

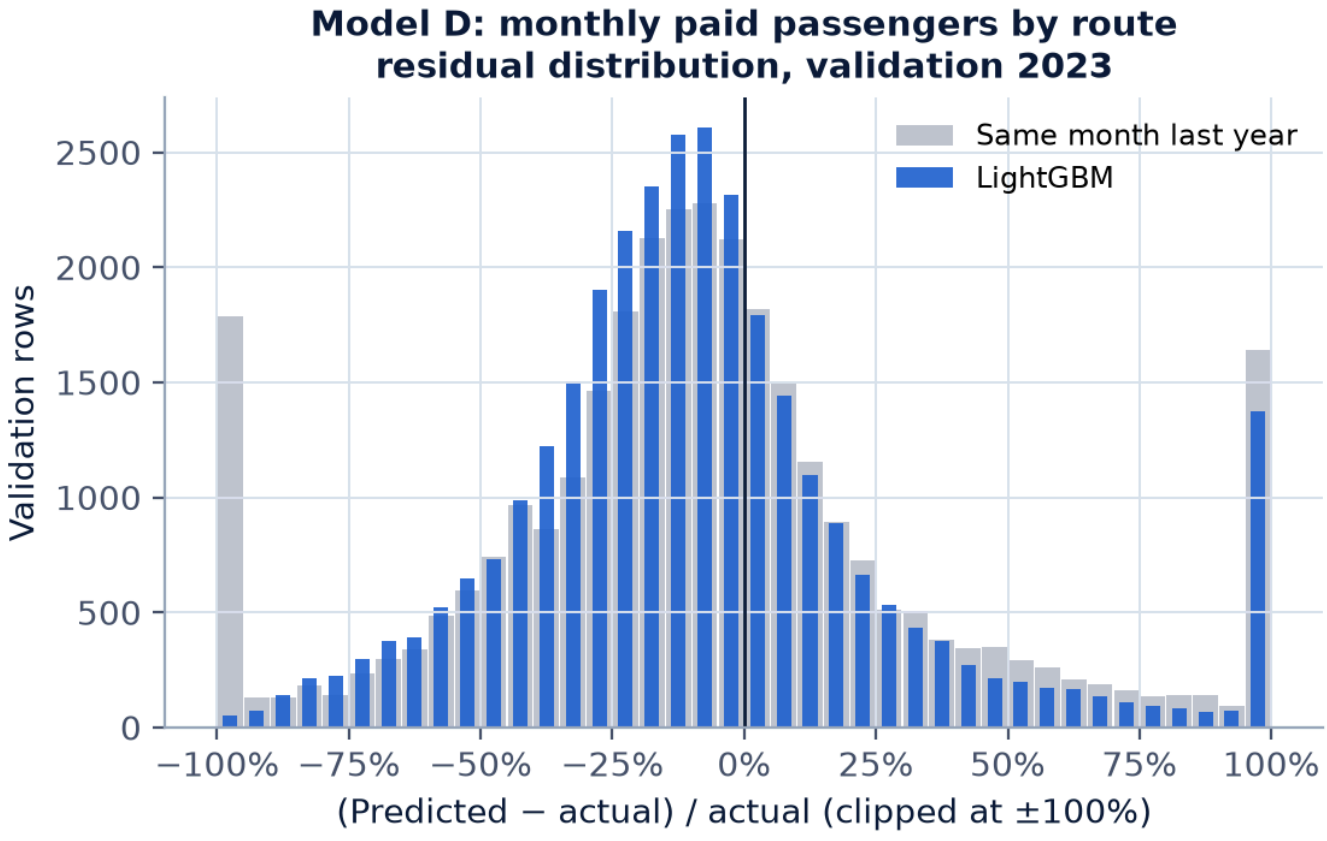




- **A:** the error is highest in November and December (12.1% and 12.8% at 1 month ahead). LightGBM beats the baseline on Saturdays and Sundays (11.6% vs 12.5%; 10.3% vs 11.6%) and is level on weekdays.
- **D:** LightGBM's monthly WAPE at 1 month ahead ranges from 14.6% (September) to 23.2% (December), below the baseline in every month.

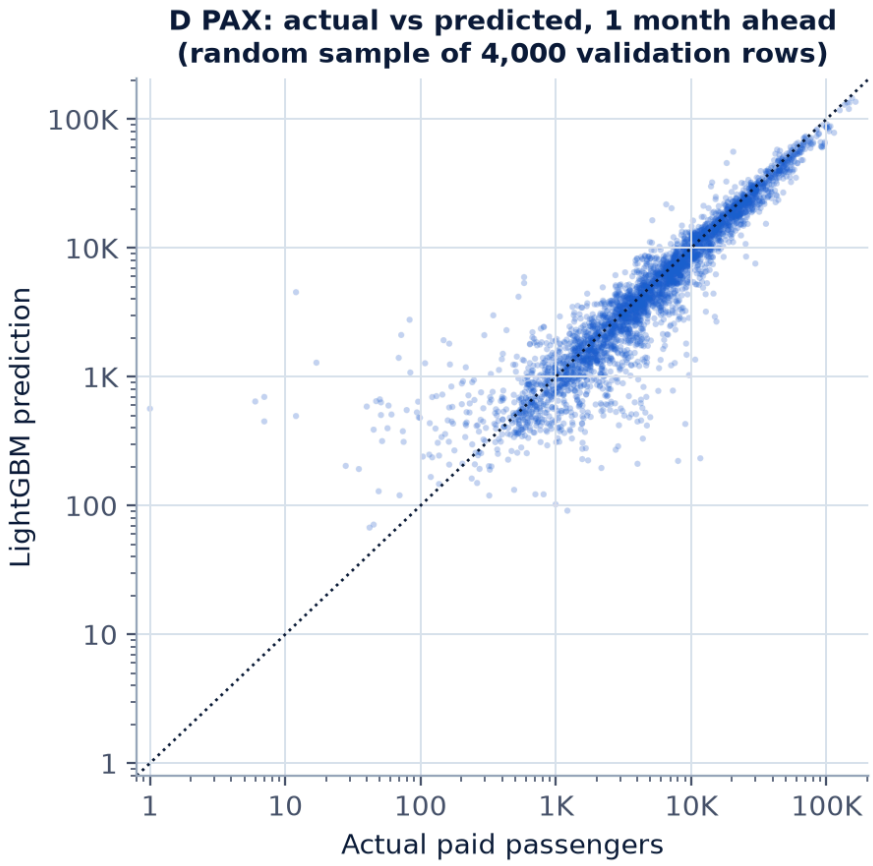
12.4 Residuals

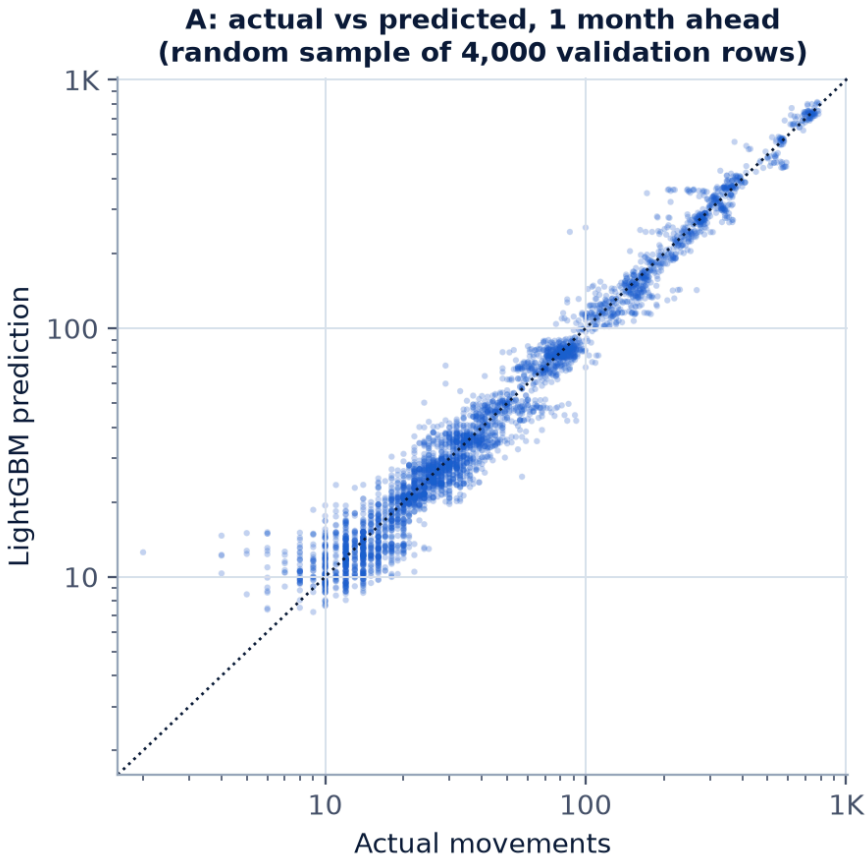




Task	Residual quantiles, LightGBM (p10 / p50 / p90)	Best baseline
A, relative	−22.6% / −2.6% / +21.7%	−20.8% / −1.3% / +26.1%
B, points	−10.2 / −1.4 / +11.9	−12.9 / −2.3 / +10.2
D pax, relative	−48.5% / −11.2% / +37.4%	−60.9% / −9.8% / +51.8%

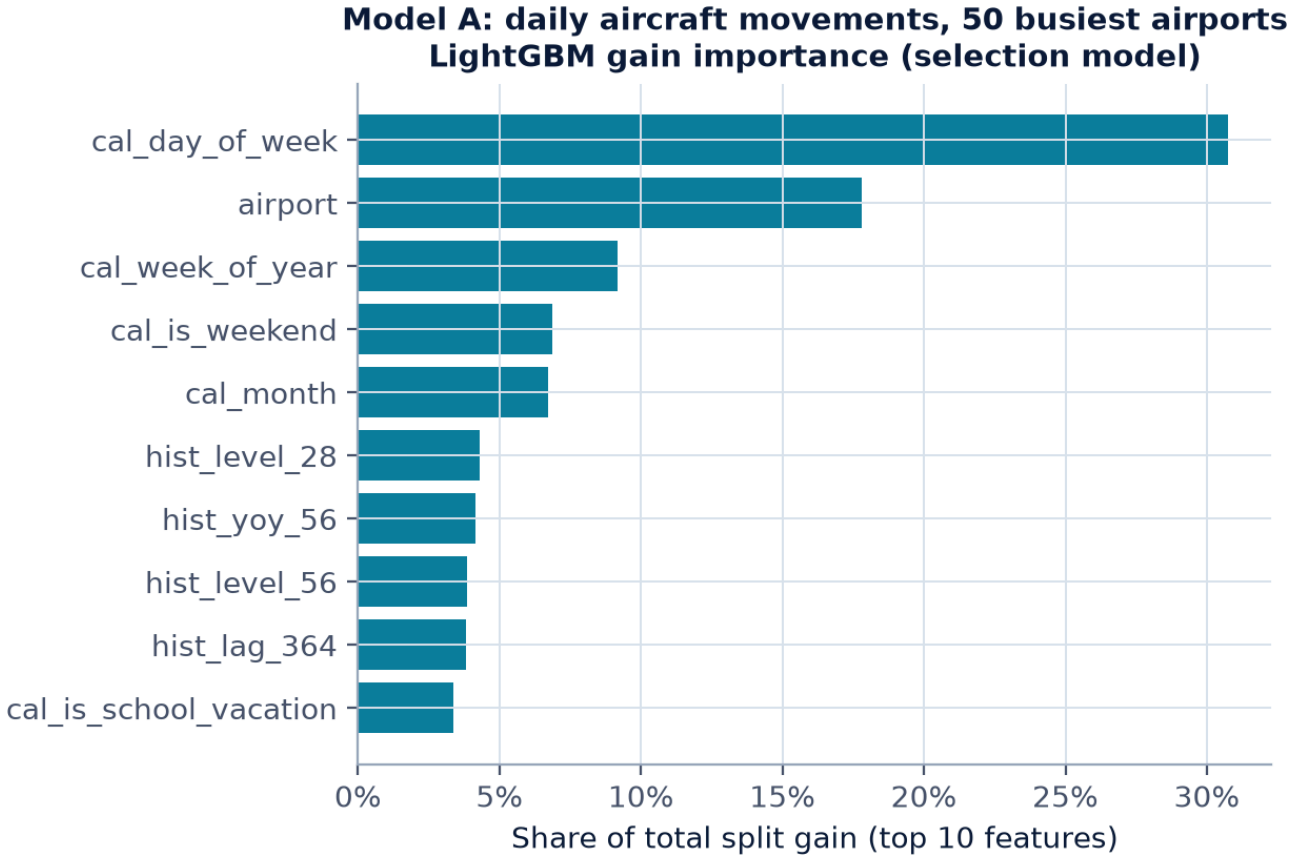
D's residuals are tighter than the baseline's but centred below zero, which is the −12% bias seen per row.

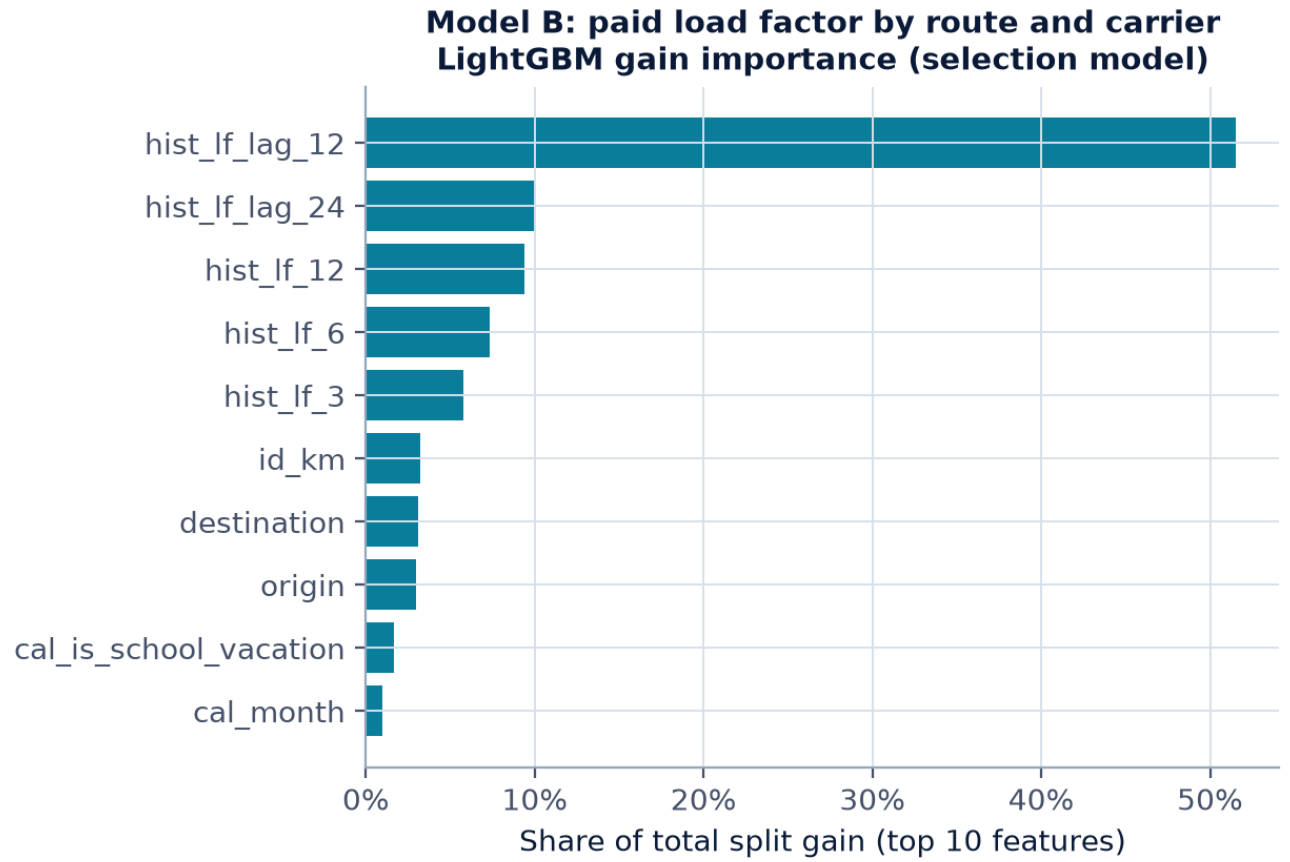


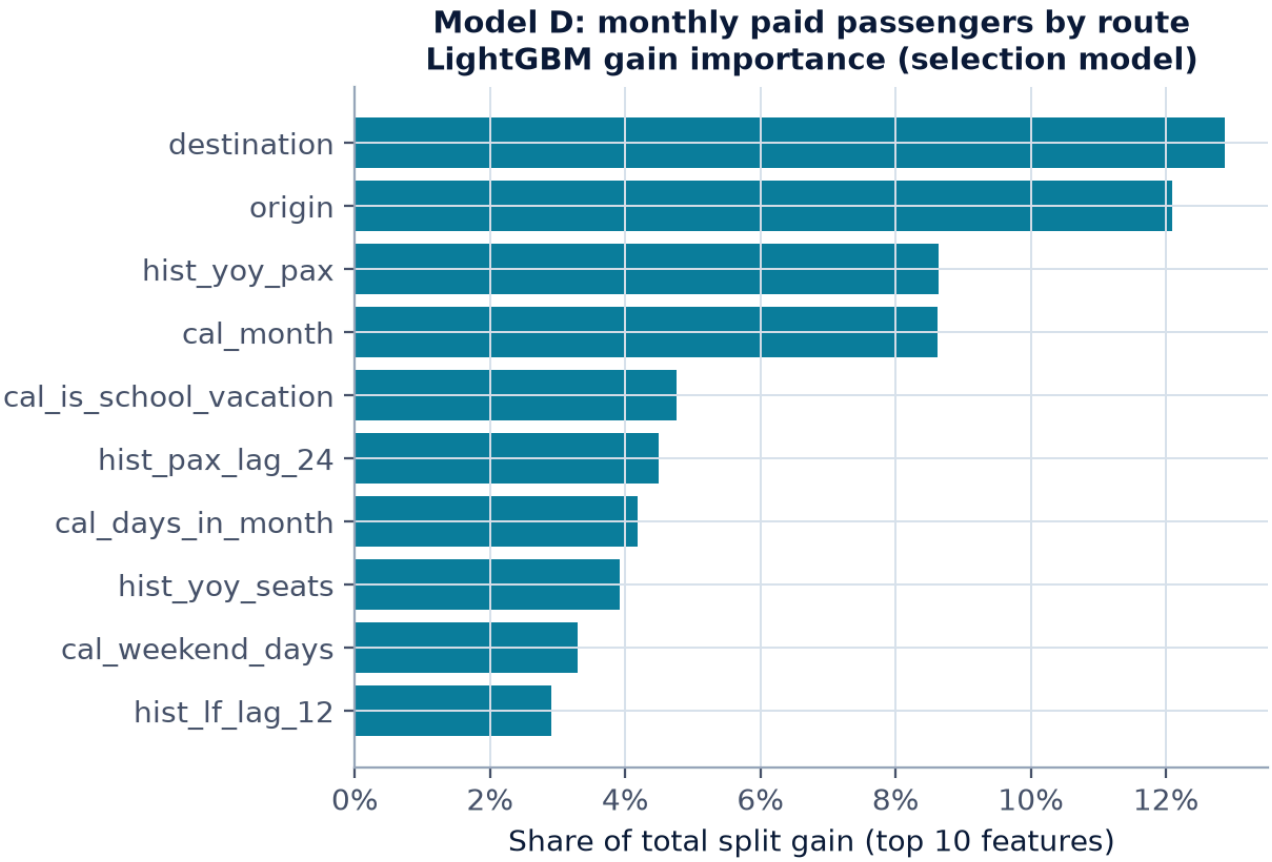


13. Explainability

The only attribution available is **LightGBM split gain** from the selection models (`reports/models/*.json`). It shows what the trees split on. It is **not** a causal effect and **not** SHAP. No SHAP analysis was run.


Model A: daily aircraft movements, 50 busiest airports
LightGBM gain importance (selection model)
cal_day_of_week
airport
cal_week_of_year
cal_is_weekend
cal_month
hist_level_28
hist_yoy_56
hist_level_56
hist_lag_364
cal_is_school_vacation
0%
5%
10%
15%
20%
25%
30%
Share of total split gain (top 10 features)


Model B: paid load factor by route and carrier
LightGBM gain importance (selection model)
hist_lf_lag_12
hist_lf_lag_24
hist_lf_12
hist_lf_6
hist_lf_3
id_km
destination
origin
cal_is_school_vacation
cal_month
0%
10%
20%
30%
40%
50%
Share of total split gain (top 10 features)



Model	Top features by gain
A	Day of week 30.8%, airport 17.8%, week of year 9.2%, weekend 6.9%, month 6.7%
B	Load factor 12 months earlier 51.5%, 24 months earlier 10.0%, 12-month mean 9.4%, 6-month mean 7.3%
D pax	Destination 12.9%, origin 12.1%, year-over-year change 8.6%, month 8.6%, school vacation 4.8%

These are consistent with the error analysis:

- **A** is a calendar problem, which is why a weekday rule competes.
- **B** leans on last year's load factor.
- **D** relies on route identity and year-over-year change, which in 2023 understated the growth.

14. Practical Applications

14.1 What the evidence supports

Use	Model	Condition
Airport staffing and slot planning, 1–3 months ahead, at major airports	**Baseline A** (recent same weekday)	About 10% WAPE. At large airports such as GRU, about 4–6%.
Load-factor expectation per route and carrier, for capacity reviews	**LightGBM B**	About 6 pts MAE, unbiased. Strongest at 6 months and on domestic routes.
Relative route demand: ranking and seasonality, 1–3 months ahead	**LightGBM D**	Shape and ranking are better than last year's values. Levels need a market-level correction.

14.2 What it does not support

- **Absolute route demand without a level correction:** Model D runs about 12% low in a growth year.
- **Six months ahead for Model D:** no better than last year.
- **Structural changes,** such as an airport losing traffic: no history-based model sees them coming.
- **Unconstrained demand or price:** the data measures passengers carried, not demand, and has no fares.

14.3 Why there is no free-input prediction playground

The models consume dozens of features computed from history at a cutoff: lags, rolling means, year-over-year change and calendar effects. A user cannot type these honestly. A form would either ask for values only known after the target month, which is leakage, or silently invent them.

The frozen `v1` models have also not been tested beyond 2023, Module A's model does not beat its baseline, and no serving path builds features for new origins.

The platform therefore shows an **interactive explorer of the verified 2023 out-of-sample predictions**, and no form that generates new numbers.

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15. Limitations & Threats to Validity

1. **One validation year, and an atypical one.** 2023 was a recovery year with +15.3% growth, and the results may not transfer to a flat year. The block bootstrap covers sampling noise within 2023, not year-to-year change.
2. **Test not run** (Section 6.4). The test period is also not fully blind (decision 3).
3. **Four absent months** in the history weaken the lags that fall on them.
4. **Revisions.** Backtests use revised data, since no historical vintages exist (decision 5). This may flatter the most recent months.
5. **Conditional evaluation in B:** routes that stopped flying are not scored.
6. **Per-leg counting** and a snapshot of the source that predates later ANAC revisions.
7. **Gain importance is not attribution.** No causal claims are made.
8. **Baselines were reported before the models** (`reports/baselines/`, `docs/features.md`). The model configuration was fixed before validation, but the baselines were known, as is standard.

---

16. Reproducibility

```
# from the repository root, with data/features/v1, models/v1 and data/analytics/v1 built locally
.venv\Scripts\python -m pip install -r research\ml-showcase\requirements.txt
.venv\Scripts\python research\ml-showcase\reproduce.py --refit   # ~40 s: metrics, hashes, in-memory refit
.venv\Scripts\python research\ml-showcase\analysis.py            # ~20 s: figures, tables, web payload
.venv\Scripts\python research\ml-showcase\build_pdf.py           # REPORT.md -> REPORT.html -> REPORT.pdf
```

Artifact	Path
Frozen models and manifests (hashes, params, git)	`models/v1/module_{a,b,d}/`
Official validation reports	`reports/models/module_{a,b,d}.json`
Saved validation predictions	`data/features/v1/module_{a,b,d}/lightgbm_predictions_validation.parquet`
Reproduction record	`research/ml-showcase/results/reproduction.json`
All tables behind this report	`research/ml-showcase/results/analysis.json`
Figures (PNG, 200 dpi, and SVG)	`research/ml-showcase/figures/`
Payload for the dashboard's ML Studio	`web/public/ml/showcase.json` (aggregates only)
Configuration	`configs/features.toml`, `configs/models.toml`
Methodology and decisions	`docs/features.md`, `docs/models.md`, `docs/decisions.md`

17. Conclusions & Future Work

Answers to the research questions:

- **RQ1:**
  - Airport movements: about 10% WAPE.
  - Seat occupancy: about 6 pts MAE.
  - Route demand: about 19% WAPE at 1 month, rising to about 27% at 6.
- **RQ2:** Machine learning beats simple baselines for **B** (small, conclusive) and **D** (clear at 1–3 months). It does not for **A**, where a weekday rule is as good.
- **RQ3:** Errors concentrate in small units, in structural breaks (SBRJ in late 2023), in international routes for B, and in a level bias for D during growth.
- **RQ4:** The models are useful for planning signals and relative comparisons, with the stated error bands. They do not replace absolute forecasts in a regime change.

Next steps, in order:

1. Recover the four absent months and retrain as `models/v2`.
2. Run the sealed 2024–2025 test once, without tuning.
3. Add a market-level correction, or the year-over-year level of the origin month, to D, and evaluate it on validation before the test.
4. Report results per year when more validation years become available.
5. Serve forecasts only after the test, with the error bands above attached.

Generated from artifacts in the ANAC Flight Intelligence repository. Data: ANAC open data, published under Brazil's access to information law.